

LangRobo Hardware Wiring Documentation v1.0

Current hardware: ESP32 DevKit V1, 2x BTS7960 (IBT-2), 4 Encoder DC Motors, 12V Battery, 65W USB Power Bank.

Power Connections

- Power Bank -> Micro USB -> ESP32
- 12V Battery (+) -> Left BTS7960 B+, Right BTS7960 B+
- 12V Battery (-) -> Left BTS7960 B-, Right BTS7960 B-, ESP32 GND (common ground)

ESP32 to Left BTS7960

BTS7960	ESP32
VCC	VIN
GND	GND
RPWM	GPIO18
LPWM	GPIO19
R_EN	GPIO21
L_EN	GPIO22
R_IS	NC
L_IS	NC

ESP32 to Right BTS7960

BTS7960	ESP32
VCC	VIN
GND	GND
RPWM	GPIO23
LPWM	GPIO5
R_EN	GPIO27
L_EN	GPIO13
R_IS	NC
L_IS	NC

Motor Connections

Left BTS7960 M+ -> Left Front Red + Left Rear Red
Left BTS7960 M- -> Left Front White + Left Rear White
Right BTS7960 M+ -> Right Front Red + Right Rear Red
Right BTS7960 M- -> Right Front White + Right Rear White

Encoder Connections

Encoder	Blue(V)	Black(G)	Green(A)	Yellow(B)
Left Front	VIN	GND	GPIO34	GPIO35

Left Rear	VIN	GND	GPIO36(VP)	GPIO39(VN)
Right Front	VIN	GND	GPIO32	GPIO33
Right Rear	VIN	GND	GPIO25	GPIO26

Important Notes

- Connect ALL grounds together (Battery -, ESP32 GND, both BTS7960 GNDs, encoder grounds).
- Leave R_IS and L_IS unconnected.
- Verify encoder supply voltage (5V vs 3.3V) from the motor documentation before final soldering.
- Motors on each side are connected in parallel to one BTS7960.